

Q12

```
1  #include<stdio.h>
2  int main()
3  {
4      char n;
5      printf("Enter a character: ");
6      scanf(" %c", &n);
7      printf("The ascii value of the character you entered is: '%d'\n", n);
8      return 0;
9  }
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Programs> cd
($?) { gcc 12th_file.c -o 12th_file } ; if ($?) { .\12th_file }
Enter a character: a
The ascii value of the character you entered is: '97'
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Programs\Lab
```

ab_2 > C 13th_file.c > main()

Q13

```
1  #include<stdio.h>
2  int main()
3  {
4      int N;
5      printf("Enter a number: ");
6      scanf("%d", &N);
7      if(N%2!=0)
8      {
9          printf("N is odd");
10     }
11     if(N%2==0 && N!=0)
12     {
13         printf("N is even\n");
14     }
15
16
17     return 0;
18 }
```

output:

```
PS C:\Users\RUTVI MAK
($?) { gcc 13th_file
Enter a number: 78
N is even
PS C:\Users\RUTVI MAK
```

```

1  #include<stdio.h>
2  int main()
3  {
4      char ch;
5      printf("enter the value of character: ");
6      scanf("%c",&ch);
7      if(ch>='A' && ch<='Z')
8      {
9          printf("The character is in upper case");
10     }
11     else if(ch>='a' && ch<='z')
12     {
13         printf("The character is in lower case");
14     }
15     else if(ch>='0' && ch<='9')
16     {
17         printf("The character is a digit");
18     }
19     else
20     {
21         printf("The character is a special character");
22     }
23
24     return 0;}
25

```

Q14

output:

```

PS C:\Users\RUTVI MAKAWANA\Desktop\VS C
($?) { gcc 14th_file.c -o 14th_file }
enter the value of character: #
The character is a special character
PS C:\Users\RUTVI MAKAWANA\Desktop\VS C

```

Q15

```
1  #include<stdio.h>
2  void main()
3  {
4      float S1,S2,S3,S4,S5;
5      float T,P;
6      printf("ENTER THE MARKS OF SUBJECT-1 OUT OF 100: ");
7      scanf("%f",&S1);
8      printf("ENTER THE MARKS OF SUBJECT-2 OUT OF 100: ");
9      scanf("%f",&S2);
10     printf("ENTER THE MARKS OF SUBJECT-3 OUT OF 100: ");
11     scanf("%f",&S3);
12     printf("ENTER THE MARKS OF SUBJECT-4 OUT OF 100: ");
13     scanf("%f",&S4);
14     printf("ENTER THE MARKS OF SUBJECT-5 OUT OF 100: ");
15     scanf("%f",&S5);
16     T = (S1+S2+S3+S4+S5);
17     P = (T/500)*100;
18     if(P>=90)
19     {
20         printf("You have secured A+ division");
21     }
22     else if(P<90&&P>=80)
23     {
24         printf("You have secured A division");
25     }
26     else if(P<80&&P>=70)
27     {
28         printf("You have secured B division");
29     }
30     else if(P<70&&P>=60)
31     {
32         printf("You have secured C division");
33     }
34     else if(P<60&&P>=50)
35     {
36         printf("You have secured D division");
37     }
38     else
39     {
40         printf("Repeating the class is mandatory");
41     }
42 }
```

output:

PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Progr

if (\$?) { .\15th_file }

ENTER THE MARKS OF SUBJECT-1 OUT OF 100: 79

ENTER THE MARKS OF SUBJECT-2 OUT OF 100: 89

ENTER THE MARKS OF SUBJECT-3 OUT OF 100: 68

ENTER THE MARKS OF SUBJECT-4 OUT OF 100: 47

ENTER THE MARKS OF SUBJECT-5 OUT OF 100: 99

You have secured B division

PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Progr

Q16

```
1  #include <stdio.h>
2
3  int main()
4  {
5      float n1, n2, n3;
6      float max;
7
8      printf("Enter a number: ");
9      scanf("%f", &n1);
10
11     printf("Enter a number: ");
12     scanf("%f", &n2);
13
14     printf("Enter a number: ");
15     scanf("%f", &n3);
16
17     if (n1 >= n2)
18     {
19         if (n1 >= n3)
20         {
21             max = n1;
22         }
23         else
24         {
25             max = n3;
26         }
27     }
28     else
29     {
30         if (n2 >= n3)
31         {
32             max = n2;
33         }
34         else
35         {
36             max = n3;
37         }
38     }
39
40     printf("The largest number is: %f", max);
41
42     return 0;
43 }
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop\VS
if ($?) { .\16th_file }
Enter a number: 8
Enter a number: 578
Enter a number: 97
The largest number is: 578.000000
PS C:\Users\RUTVI MAKAWANA\Desktop\VS
```

```
1  #include<stdio.h>
2  int main()
3  {
4      float n1,n2,n3;
5      printf("Enter a number: ");
6      scanf("%f",&n1);
7      printf("Enter a number: ");
8      scanf("%f",&n2);
9      printf("Enter a number: ");
10     scanf("%f",&n3);
11     if(n1>n2 && n1>n3)
12     {
13         printf("n1 is the largest number");
14     }
15     else if(n2>n1 && n2>n3)
16     {
17         printf("n2 is the largest number");
18     }
19     else
20     {
21         printf("n3 is the largest number");
22     }
23 }
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop>
if ($?) { .\17th_file }
Enter a number: 6
Enter a number: 87
Enter a number: 999
n3 is the largest number
PS C:\Users\RUTVI MAKAWANA\Desktop>
```

Q18

```
1  #include<stdio.h>
2  int main()
3  {
4      float n1,n2,n3;
5      float max;
6      printf("Enter a number: ");
7      scanf("%f",&n1);
8      printf("Enter a number: ");
9      scanf("%f",&n2);
10     printf("Enter a number: ");
11     scanf("%f",&n3);
12     max=(n1>n2)?((n1>n3)?n1:n3):((n2>n3)?n2:n3);
13     printf("The largest number is: %f",max);
14     return 0;
15 }
16
```

output:

PS C:\Users\RUTVI MAKAWANA\Desktop\VS

if (\$?) { .\18th_file }

Enter a number: 58

Enter a number: 89

Enter a number: 67

The largest number is: 89.000000

PS C:\Users\RUTVI MAKAWANA\Desktop\VS

Q19

```
1  #include<stdio.h>
2  int main(int a, int b)
3  {
4      char ch;
5      printf("enter the value of character: ");
6      scanf("%c",&ch);
7      if(ch>='a' && ch<='z')
8      {
9          printf("The character is in lower case");
10     }
11     else
12     {
13         printf("The character is not lower case");
14     }
15     return 0;
16 }
```

Output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop\W
if ($?) { .\19th_file }
enter the value of character: H
The character is not lower case
PS C:\Users\RUTVI MAKAWANA\Desktop\W
```

Q20

```
1  #include<stdio.h>
2  int main()
3  {
4      int a,b,c;
5      printf("Enter two numbers: ");
6      scanf("%d%d",&a,&b);
7      printf("Enter 1 if you want to add the numbers\n ");
8      printf("Enter 2 if you want to subtarct the numbers\n ");
9      printf("Enter 3 if you want to multiply the numbers\n ");
10     printf("Enter 4 if you want to divide a by b\n ");
11     printf("Enter a choice: ");
12     scanf("%d",&c);
13     switch(c)
14     {
15         case 1:
16             printf("Addition: %d",a+b);
17             break;
18
19         case 2:
20             printf("Subtraction: %d",a-b);
21             break;
22
23         case 3:
24             printf("Multipilcation: %d",a*b);
25             break;
26
27         case 4:
28             printf("Division: %d",a/b);
29             break;
30
31         default:
32             printf("Invalid choice");
33
34     }
35     return 0;
36 }
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Pro
if ($?) { .\20th_file }
```

Enter two numbers: 56

90

Enter 1 if you want to add the numbers

Enter 2 if you want to subtarct the numbers

Enter 3 if you want to multiply the numbers

Enter 4 if you want to divide a by b

Enter a choice: 2

Subtraction: -34

```
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Pro
```

Q21

```
1  #include <stdio.h>
2  int main()
3  {
4      int a, b;
5      char op;
6
7      printf("Enter two numbers: ");
8      scanf("%d %d", &a, &b);
9
10     printf("Enter operation (+, -, *, /, %%): ");
11     scanf(" %c", &op);
12
13     switch(op)
14     {
15         case '+':
16             printf("Result = %d", a + b);
17             break;
18         case '-':
19             printf("Result = %d", a - b);
20             break;
21         case '*':
22             printf("Result = %d", a * b);
23             break;
24         case '/':
25             printf("Result = %d", a / b);
26             break;
27         case '%':
28             printf("Result = %d", a % b);
29             break;
30         default:
31             printf("Invalid operator");
32     }
33
34     return 0;
35 }
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop\VS
if ($?) { .\21st_file }
Enter two numbers: 7
9
Enter operation (+, -, *, /, %): +
Result = 16
PS C:\Users\RUTVI MAKAWANA\Desktop\VS
```

```
1  #include<stdio.h>
2  int main()
3  {
4      float n1;
5      printf("Enter a number: ");
6      scanf("%f", &n1);
7      float factorial = 1;
8      for(int i=1; i<=n1; i++)
9      {
10         factorial *= i;
11     }
12     printf("Factorial: %f\n", factorial);
13     return 0;
14 }
```

output:

```
PS C:\Users\RUTVI MAKAWANA
if ($?) { .\22nd_file }
Enter a number: 6
Factorial: 720.000000
PS C:\Users\RUTVI MAKAWANA
```

```
1  #include<stdio.h>
2  int main()
3  {
4      int base, exponent;
5      int result=1;
6      printf("Enter a base number: ");
7      scanf("%d", &base);
8      printf("Enter an exponent: ");
9      scanf("%d", &exponent);
10     for (int i=1; i<=exponent; i++)
11     {
12         result *= base;
13     }
14     printf("%d",result);
15     return 0;
16 }
```

output:

PS C:\Users\RUTVI MAKAWANA\D

if (\$?) { .\23rd_file }

Enter a base number: 5

Enter an exponent: 2

25

PS C:\Users\RUTVI MAKAWANA\D